

Auditable Without Transcripts: Aggregate-Only Reproducibility for Sensitive Speech AI Evidence

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Abstract

Sensitive speech AI studies increasingly produce evidence from audio, transcripts, model hypotheses, reviewer judgments, runtime logs, and operational decisions. Responsible AI documentation and risk-management practice encourage traceable evaluation, while sensitive speech settings require a release boundary that protects transcript-bearing material. This paper introduces aggregate-only reproducibility as a reviewer-auditable evidence architecture for that setting. The framework turns sensitive evaluation into a structured claim-evidence package: public aggregate artifacts, local-only transcript-bearing evidence, artifact manifests, validation gates, operation records, privacy classes, and explicit release-boundary decisions.

We develop the framework through a CDS-ASR case study in high-stakes speech-to-decision evaluation. The case contains four distinct evidence layers: a 258-row ASR test split for model-comparison context, selected-300 provenance outputs for enriched audit-surface construction, a completed 300-row dual-reviewer human audit with 900 model-level assessments per reviewer and Cohen’s kappa agreement summaries, and aggregate policy replay for recovery analysis. Instead of releasing raw speech or transcript-bearing row content, the case provides reviewer-visible auditability through aggregate tables, generated figures, claim registries, artifact manifests, agreement summaries, validation checks, and release-boundary records.

The contribution is claim-evidence stewardship for sensitive speech AI. Aggregate-only reproducibility gives reviewers a concrete way to inspect artifact completeness, metric definitions, validation gates, agreement evidence, figure regeneration, and claim-scope alignment while transcript-bearing rows remain under governed local control. It complements row-level reproduction with a privacy-preserving audit mode for studies whose evidence can be made inspectable without public transcript release.

Keywords: trustworthy AI; reproducible evaluation; sensitive speech data; artifact manifest; claim-evidence alignment; responsible AI audit.

1 Introduction

Sensitive speech AI research sits between two necessary commitments: empirical claims should be traceable, and transcript-bearing evidence should remain governed when it contains private operational speech content. Raw audio, transcripts, row identifiers, reviewer notes, response sheets, model hypotheses, and transcript-bearing runtime logs can carry contextual details that are subject to institutional, contractual, ethical, or safety controls. Reviewers still need to inspect whether the paper’s claims follow from the evidence.

This paper makes reviewer auditability the central design target. Aggregate-only reproducibility exposes the evidence that can be safely shared—aggregate tables, figure inputs, artifact manifests, validation gates, operation records, claim registries, agreement summaries, and release-boundary

decisions—while keeping transcript-bearing evidence inside the declared local governance boundary. The result is an inspectable evidence package rather than an opaque privacy caveat.

The paper uses the CDS-ASR project as a case study. The technical details of CDS-ASR and CEIS belong to the companion speech-evaluation manuscript. Here, CDS-ASR demonstrates how a sensitive speech AI study can turn model-comparison context, selected-300 provenance, completed dual-reviewer human audit evidence, agreement statistics, and aggregate policy replay into a reviewer-auditable package without public transcript release.

2 Background

Reproducibility in AI evaluation. AI evaluation commonly relies on public datasets, executable code, fixed metrics, and artifact checklists. Sensitive speech settings add a governance layer because the evaluation evidence may include personally revealing acoustic or transcript-bearing content. Aggregate-only reproducibility keeps the audit goal intact by making metric definitions, manifests, validation records, and claim mappings visible even when transcript-bearing rows stay local.

Sensitive speech data. Speech data can contain identity cues, operational details, medical or financial information, emotional content, and contextual signals that remain sensitive even after transcription. Releasing row-level examples can therefore create privacy and governance risks.

Claim-evidence alignment. For sensitive AI evaluation, the paper claim should specify its evidence layer, statistic, scope, validation status, privacy class, and next validation path. This turns scope control into a claim-evidence alignment discipline. The framework is compatible with documentation-oriented responsible-AI practice such as dataset documentation [1] and risk-management framing [2], and it gives reviewers a concrete way to audit what each claim can support.

Audit trails and operation records. Operation records preserve process-level reproducibility: what generator or validator produced an artifact, from which source inputs, under which version and environment, with which checksum. They do not expose local-only content, but they make the aggregate evidence trail inspectable.

3 Aggregate-Only Reproducibility Framework

We propose a five-layer framework for sensitive speech AI evaluation.

3.1 Layer 1: Evidence Boundary

The evidence boundary defines which materials are public aggregate artifacts and which remain local-only sensitive evidence. Figure 2 maps the boundary.

3.2 Layer 2: Aggregate Artifact Layer

The aggregate artifact layer releases transcript-free summary tables, metric outputs, validation summaries, manifest rows, figure inputs, agreement summaries, and completion checks. Transcript-bearing rows, identifiers, reviewer notes, response sheets, model hypotheses, and transcript-bearing logs remain governed local evidence.

Five-layer aggregate-only reproducibility framework

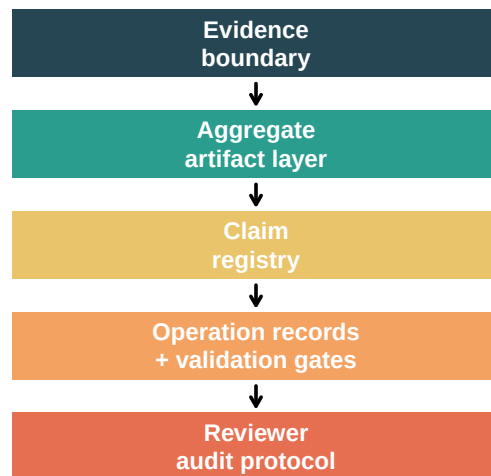


Figure 1: Five-layer aggregate-only reproducibility framework. The figure is generated by `paper_4b_aggregate_reproducibility/r/build_fig1_framework_layers.R` as a conceptual framework derived from the manuscript protocol.

Evidence boundary map

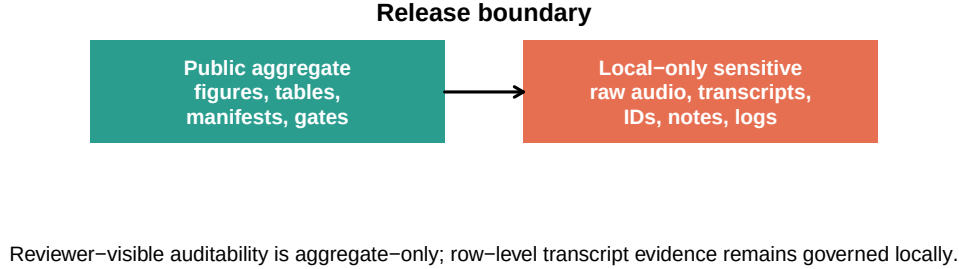


Figure 2: Evidence boundary map separating public aggregate artifacts from local-only sensitive evidence. Generated by `build_fig2_evidence_boundary.R`.

3.3 Layer 3: Claim Registry

The claim registry maps every paper-facing claim to artifact path, statistic, scope, limitation, privacy class, and validation status. Figure 3 shows the claim-to-evidence flow.

3.4 Layer 4: Operation Records And Validation Gates

Operation records and validation gates provide process-level audit trails without exposing sensitive content. They include artifact manifests, evidence-chain readiness checks, consistency checks, publishable-evidence summaries, and leak checks.

3.5 Layer 5: Reviewer Audit Protocol

The reviewer audit protocol specifies what reviewers can inspect: artifact completeness, claim-evidence alignment, consistency checks, metric definitions, agreement summaries, scope boundaries, and R-based figure regeneration. Row-level qualitative interpretation remains a governed local validation layer when public release would expose sensitive transcript-bearing content.

4 Case Study: CDS-ASR

CDS-ASR is used here only as a case study for aggregate-only reproducibility. The companion Paper 4-a carries the speech-evaluation method and CEIS technical claims.

The case has four evidence layers: a 258-row ASR split, selected-300 enriched high-stakes provenance outputs, a completed 300-row dual-reviewer human audit, and 900 model-row assessments per reviewer. Figure 4 shows the case-study evidence ladder.

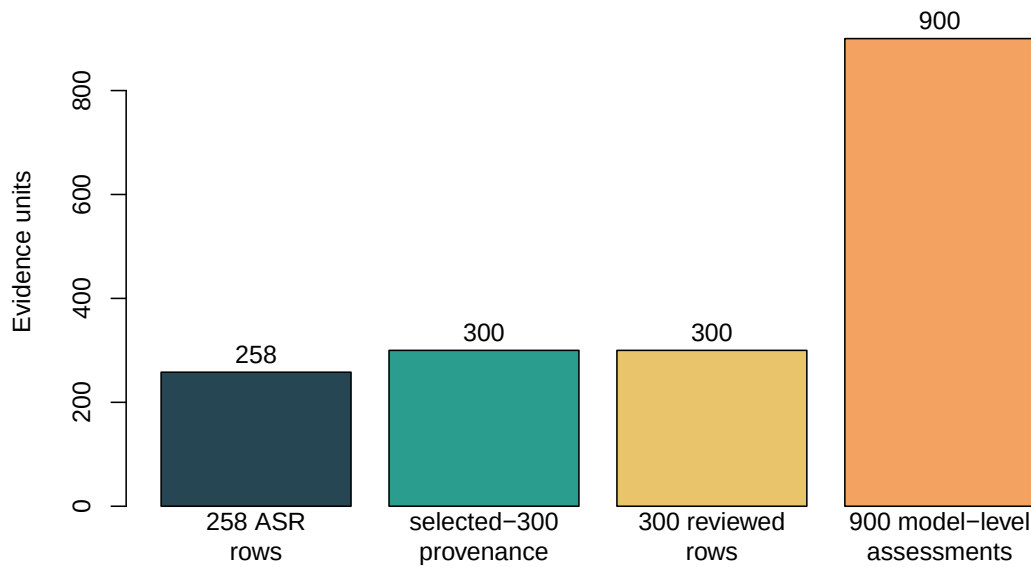
Claim registry flow



Every claim is routed through aggregate evidence and an explicit boundary.

Figure 3: Claim registry flow from paper claim to artifact, statistic, scope, limitation, privacy class, and validation status. Generated by `build_fig3_claim_registry_flow.R`.

CDS-ASR case-study evidence ladder



Case evidence is aggregate-only; selected-300 is enriched and dual-reviewer agreement is reported.

Figure 4: CDS-ASR case-study evidence ladder. Generated by `paper_4b_aggregate_reproducibility/r/build_fig5_case_study_evidence_ladder.R`.

Predictor comparison, reviewer agreement, and policy replay appear as aggregate outputs. Raw audio, raw transcripts, row identifiers, reviewer notes, response sheets, model hypotheses, and transcript-bearing logs remain local-only. This makes the case appropriate for demonstrating reviewer-auditable aggregate evidence while preserving sensitive material.

5 Artifact And Claim-Evidence Protocol

The protocol combines artifact manifests, claim registries, evidence-chain consistency summaries, publishable-evidence completion summaries, roadmap completion summaries, operation records, validation gates, and privacy classes.

Table 1: Public, aggregate-only, and local-only evidence classes.

Class	Examples	Reviewer use
Public aggregate	Summary tables, generated figures, validation summaries	Check reported aggregate claims
Public source/config	Figure scripts, manifests, metric definitions	Regenerate public figures and inspect definitions
Controlled aggregate supplement	Extended aggregate tables and sensitivity summaries	Inspect scoped uncertainty and consistency
Local-only sensitive	Raw audio, transcripts, row IDs, reviewer notes, response sheets, hypotheses	Not public; governed locally or institutionally

Table 2: Claim-evidence alignment schema.

Field	Purpose
Claim	Paper-facing statement being supported
Artifact path	Aggregate artifact used as evidence
Statistic	Reported value or validation status
Scope	Evidence layer and population boundary
Limitation	Claim-evidence alignment boundary
Privacy class	Public aggregate, controlled aggregate, or local-only
Validation status	Gate, check, or operation record supporting use

The core case-study artifacts include `70_experiments/runs/postdoc_evidence_chain_2026_05_25/artifact_manifest.tsv`, `70_experiments/runs/postdoc_evidence_chain_2026_05_25/claim_registry.tsv`, `70_experiments/runs/postdoc_evidence_chain_2026_05_25/evidence_chain_consistency_summary.json`, `70_experiments/runs/postdoc_evidence_chain_2026_05_25/publishable_evidence_completion_summary.json`, and `70_experiments/runs/postdoc_evidence_chain_2026_05_25/postdoc_roadmap_completion_summary.json`.

6 Reviewer-Auditable Boundary

Reviewer-visible auditability complements row-level reproducibility with a release mode designed for sensitive speech evidence. Reviewers can inspect aggregate artifacts, figures, tables, manifests,

Table 3: Artifact manifest fields and purpose.

Field	Purpose
Path	Locates the aggregate artifact
Role	Explains paper-facing function
Generator	Identifies the script or process that produced it
Source inputs	Defines upstream aggregate evidence
Checksum	Supports integrity checks
Source commit	Links artifact to repository state
Environment	Records runtime and package context
Privacy class	Confirms release boundary

validation gates, claim registries, agreement summaries, completion checks, and release-boundary statements. Raw transcript cases remain part of the governed local evidence layer.

Reviewer auditability map

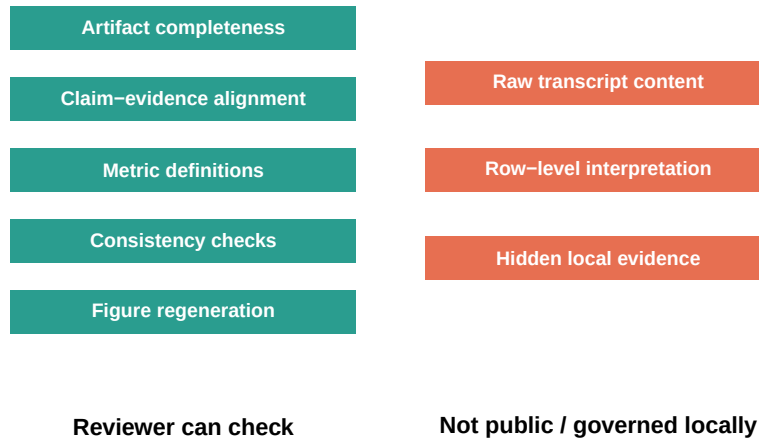


Figure 5: Reviewer auditability map showing what can be checked and what remains local-only or institutionally governed. Generated by `build_fig4_artifact_audit_map.R`.

7 Discussion

Aggregate-only reproducibility is useful for sensitive speech AI domains where public row-level release requires strong governance: medical speech, fraud calls, psychotherapy speech, law-enforcement-adjacent speech data, and high-stakes conversational AI evaluation. The framework gives reviewers a structured way to inspect the public evidence layer while transcript-bearing materials remain under declared controls.

Table 4: Reviewer audit checklist.

Reviewer check	Public evidence	Boundary
Artifact completeness	Manifests and aggregate package list	Does not expose raw rows
Claim-evidence alignment	Claim registry	Claims remain scoped
Metric definitions	Method summary and metric audit	Raw transcripts not released
Reviewer agreement	Cohen’s kappa summary	Transcript-bearing sheets remain local
Consistency checks	Evidence-chain summaries	Aggregate validation only
Figure regeneration	R scripts and aggregate inputs	R scripts cannot access sensitive inputs

Table 5: CDS-ASR case-study claim boundary table.

Case evidence	What it supports	Boundary
258 ASR rows	Model-comparison context	Not deployment safety evidence
selected-300 provenance	Enriched audit surface	Not prevalence-preserving
300 reviewed rows	Human-reviewed audit layer	Dual-reviewer agreement reported
900 model assessments per reviewer	Model-level audit evidence	Clustered within selected rows
Aggregate replay	Policy-replay auditability	Retrospective, not live deployment

The framework clarifies the positive audit surface. It supports claim-evidence alignment, artifact completeness checks, metric-definition checks, agreement inspection, consistency checks, and figure regeneration from aggregate inputs. Qualitative transcript review becomes a local governed validation layer rather than a missing public artifact.

8 Scope, Ethics, And Validation Path

Aggregate-only reproducibility is a scoped audit mode for sensitive evidence. It supports reviewer inspection of claim-evidence alignment, artifact completeness, metric definitions, agreement summaries, consistency checks, and figure regeneration from aggregate inputs. Row-level qualitative re-analysis remains a governed local validation layer when public release would expose sensitive speech or transcript content. The CDS-ASR case study demonstrates the framework in one high-stakes speech-to-decision setting; broader validation can extend it across additional domains, reviewer teams, and release policies.

The framework works alongside institutional review, data-use agreements, retention policy, encryption, access control, and deployment governance. Its contribution is a claim-evidence governance structure for reviewer-auditable aggregate evidence under sensitive-data constraints.

A Appendix: Aggregate Artifacts

The public package should include aggregate artifacts only. Transcript-bearing materials remain local-only.

- Artifact manifest: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/artifact_manifest.tsv`
- Claim registry: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/claim_registry.tsv`
- Evidence-chain consistency: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/evidence_chain_consistency_summary.json`
- Publishable evidence completion: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/publishable_evidence_completion_summary.json`
- Roadmap completion: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/postdoc_roadmap_completion_summary.json`
- Paper artifact manifest: `80_semantic_risk_asr/paper/artifact_manifest.tsv`

References

- [1] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daume III, and Kate Crawford. Datasheets for datasets, 2018. Documentation background.
- [2] National Institute of Standards and Technology. Artificial intelligence risk management framework, 2023. Governance context; not cited as legal compliance evidence.